coordinate roles, avoid collisions, and adapt to dynamic battlefield conditions. The swarm can disperse, regroup, and execute multi-axis maneuvers, demonstrating scalable and resilient cooperative autonomy.

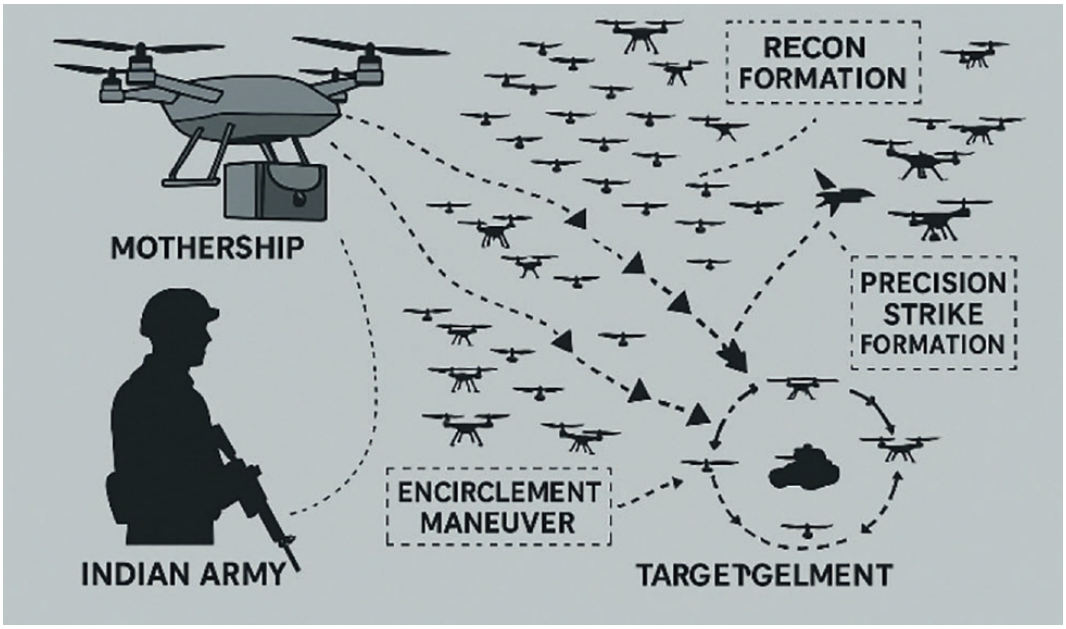


Figure 9.4 Indian Army – SWARM-100 autonomous drone swarm demonstration. ⏎

9.8.5 United States – Perdix Micro-Drone Swarm

The Perdix micro-drone program developed under US defense research initiatives served as an early and influential demonstration of practical collective autonomy in airborne swarms. Conceived as a low-cost, expendable micro-UAV that could be mass-launched from platforms such as fighter aircraft and tactical vehicles, Perdix highlighted key innovations in distributed decision-making, decentralized sensing, and fault-tolerant coordination. In public demonstrations, dozens to over a hundred Perdix units were released in mid-air and immediately organized into coordinated formations without centralized control, sharing sensor data, self-assigning roles, and dynamically reconfiguring when individual units were removed or failed.